

MATHEMATICS Compulsory Part
PAPER 2

11.30 am – 12.45 pm (1¼ hours)

INSTRUCTIONS

- Label 1 Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.
- Label 2 When told to open this book, you should check that all the questions are there. Look for the words 'END OF PAPER' after the last question.
- Label 3 All questions carry equal marks.
- Label 4 **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
- Label 5 You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
- Label 6 No marks will be deducted for wrong answers.

There are 30 questions in Section A and 15 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale.
Choose the best answer for each question.

Section A

Label 1 $\frac{(2x^4)^3}{2x^5} =$

A $3x^7$

B $3x^7$

C $4x^7$

D $4x^{59}$

Label 2 $(4x + y)^2 - (4x - y)^2 =$

A 0

B $2y^2$

C $8xy$

D $16xy$

Label 3 If p and q are constants such that $x^2 + p \equiv (x + 2)(x + q) + 10$, then $p =$

A -4

B -2

C 6

D 10

Label 4 If k is a constant such that $x^3 + 4x^2 + kx - 12$ is divisible by $x + 3$, then $k =$

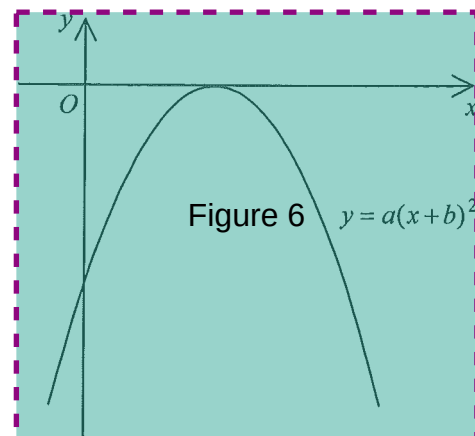
- A. -25
- B. -1
- C. 1
- D. 17

Label 5 If $m + 2n + 6 = 2m - n = 7$, then $n =$

- A. -4
- B. -1
- C. 1
- D. 1

Label 6 The figure shows the graph of $y = a(x + b)^2$, where a and b are constants. Which of the following is true?

- A. $a > 0$ and $b > 0$
- B. $a > 0$ and $b < 0$
- C. $a < 0$ and $b > 0$
- D. $a < 0$ and $b < 0$



Label 7 The solution of $15 + 4x < 3$ or $9 - 2x > 1$ is

- A. $x < -3$
- B. $x > -3$
- C. $x < 4$
- D. $x > 4$

Label 8 In a company, 37.5% of the employees are female. If 60% of the male employees and 80% of the female employees are married, then the percentage of married employees in the company is

- A. 32.5%
- B. 45%
- C. 55%
- D. 67.5%

Label 9 If x and y are non-zero numbers such that $\frac{6x+5y}{3y-2x} = 7$, then $x : y =$

- A. 4:5
- B. 4:13
- C. 5:4
- D. 13:4

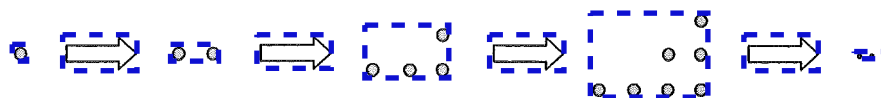
Label 10 It is given that y partly varies directly as x^2 and partly varies inversely as x . When $x = 1$, $y = -4$ and when $x = 2$, $y = 5$. When $x = -2$, $y =$

- A. -11
- B. -5
- C. 5
- D. 11

Label 11 Mary performs a typing task for 7 hours. Her average typing speeds for the first 3 hours and the last 4 hours are 63 words per minute and 56 words per minute respectively. Find her average typing speed for the 7 hours.

- A. 17 words per minute
- B. 35 words per minute
- C. 59 words per minute
- D. 60 words per minute

Label 12 In the figure, the 1st pattern consists of 1 dot. For any positive integer n , the $(n+1)$ th pattern is formed by adding n dots to the n th pattern. Find the number of dots in the 8th pattern.



- A. 22
- B. 29
- C. 36
- D. 37

Label 13 0.0322515 =

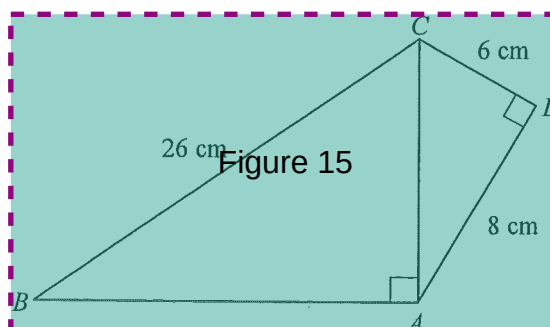
- A. 0.032 (correct to 3 significant figures)
- B. 0.0322 (correct to 4 decimal places)
- C. 0.03225 (correct to 5 significant figures)
- D. 0.032252 (correct to 6 decimal places)

Label 14 The length of a piece of thin string is measured as 25 m correct to the nearest m. If the string is cut into n pieces such that the length of each piece is measured as 5 cm correct to the nearest cm, find the greatest possible value of n .

- A. 445
- B. 565
- C. 567
- D. 650

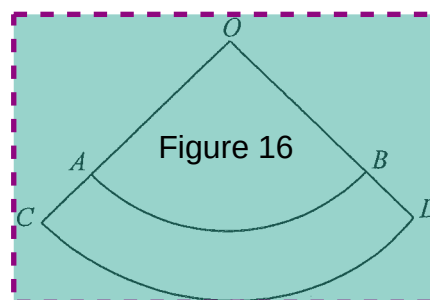
Label 15 In the figure, the area of quadrilateral $ABCD$ is

- A. 144 cm^2
- B. 160 cm^2
- C. 178 cm^2
- D. 288 cm^2



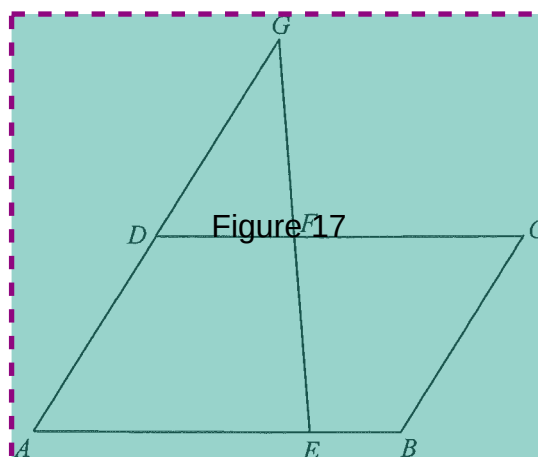
Label 16 In the figure, OAB and OCD are sectors with centre O . In $\widehat{AB} = 12\pi$ cm, $\widehat{CD} = 16\pi$ cm and $OA = 30$ cm, then $AC =$

- A! 5 cm
- B! 10 cm
- C! 20 cm
- D! 40 cm



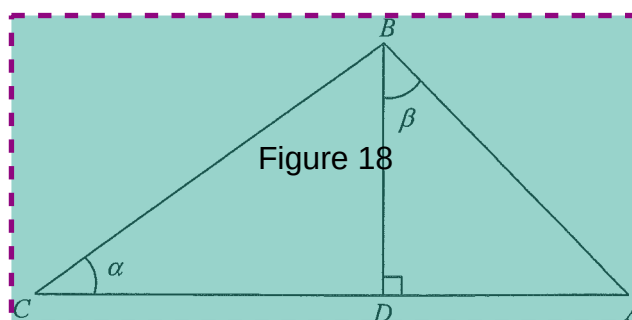
Label 17 In the figure, $ABCD$ is a parallelogram. E and F are points lying on AB and CD respectively. AD produced and EF produced meet at G . It is given that $DF : FC = 3 : 4$ and $AD : DG = 1 : 1$. If the area of $\triangle DFG$ is 3 cm^2 , then the area of the parallelogram $ABCD$ is

- A! 12 cm^2
- B! 14 cm^2
- C! 18 cm^2
- D! 21 cm^2



Label 18 In the figure, D is a point lying on AC such that BD is perpendicular to AC . If $BC = \ell$, then $AB =$

- A! $\frac{\ell \sin \alpha}{\cos \beta}$
- B! $\frac{\ell \sin \beta}{\cos \alpha}$
- C! $\frac{\ell \cos \alpha}{\sin \beta}$
- D! $\frac{\ell \cos \beta}{\sin \alpha}$



Label 19 $\frac{\cos 60^\circ}{1 - \cos(90^\circ - \theta)} + \frac{\cos 240^\circ}{1 - \cos(270^\circ - \theta)} =$

A. $\frac{1}{\cos^2 \theta}$

B. $\frac{\cos \theta}{\tan \theta}$

C. $\frac{\tan \theta}{\cos \theta}$

D. $\frac{1}{\cos \theta \tan \theta}$

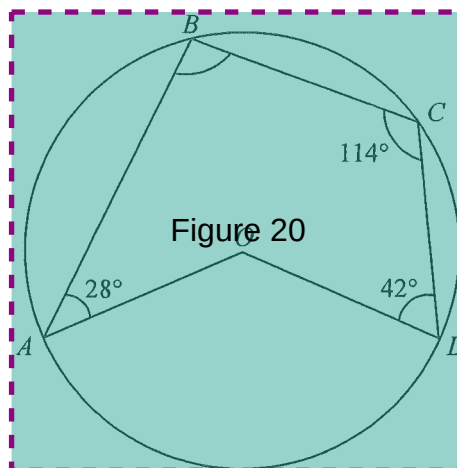
Label 20 In the figure, O is the centre of the circle $ABCD$. If $\angle BAO = 28^\circ$, $\angle BCD = 114^\circ$ and $\angle CDO = 42^\circ$, then $\angle ABC =$

A. 90°

B. 96°

C. 100°

D. 138°



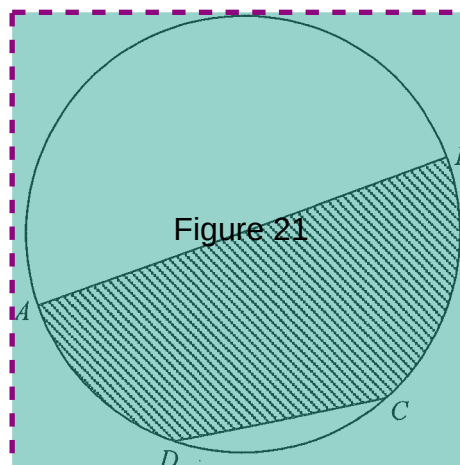
Label 21 In the figure, AB is a diameter of the circle $ABCD$. If $AB = 12$ cm and $CD = 6$ cm, then the area of the shaded region is

A. $(12\pi - 9)\text{ cm}^2$

B. $(12\pi + 9)\text{ cm}^2$

C. $(12\pi - 9\sqrt{3})\text{ cm}^2$

D. $(12\pi + 9\sqrt{3})\text{ cm}^2$



Label 22 Which of the following statements about a regular 12-sided polygon are true?

- I. Each exterior angle is 30° .
- II. Each interior angle is 150° .
- III. The number of axes of reflectional symmetry is 6.

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

Label 23 The rectangular coordinates of the point P are $(-3, -3\sqrt{3})$. If P is rotated anticlockwise about the origin through 90° , then the polar coordinates of its image are

- A. $(3, 150^\circ)$
- B. $(3, 330^\circ)$
- C. $(6, 150^\circ)$
- D. $(6, 330^\circ)$

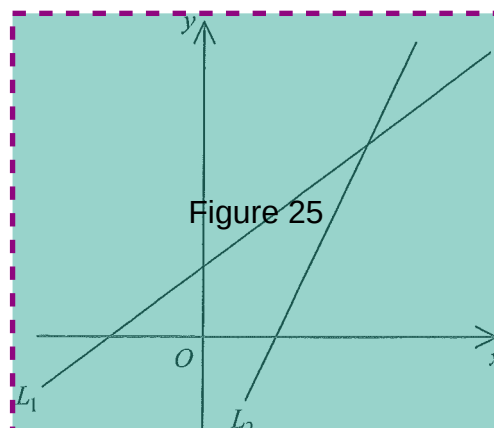
Label 24 If P is a moving point in the rectangular coordinate plane such that the distance between P and the point $(20, 12)$ is equal to 5, then the locus of P is a

- A. circle
- B. square
- C. parabola
- D. triangle

Label 25 In the figure, the equations of the straight lines L_1 and L_2 are $ax + y = b$ and $cx + y = d$ respectively. Which of the following are true?

- I. $a < 0$
- II. $a < c$
- III. $b > d$
- IV. $ad > bc$

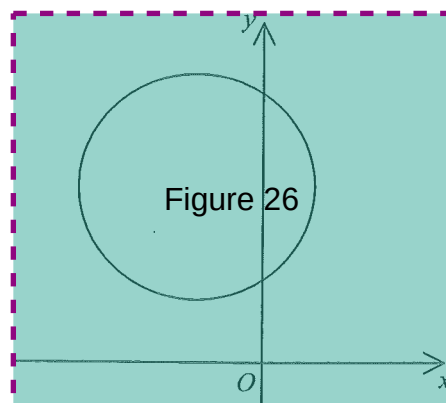
- A. I, II and III only
- B. I, II and IV only
- C. I, III and IV only
- D. II, III and IV only



Label 26 In the figure, the radius of the circle and the coordinates of the centre are r and (h, k) respectively. Which of the following are true?

- I. $h + k > 0$
- II. $r - h > 0$
- III. $r - k > 0$

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III



Label 27 $9\star\diamond$ is a 3-digit number, where $\star$ and $\diamond$ are integers from 0 to 9 inclusive. Find the probability that the 3-digit number is divisible by 5.

- A. $\frac{1}{5}$
- B. $\frac{7}{33}$
- C. $\frac{20}{99}$
- D. $\frac{19}{100}$

Label 28 The stem-and-leaf diagram below shows the distribution of the ages of a group of members in a recreational centre.

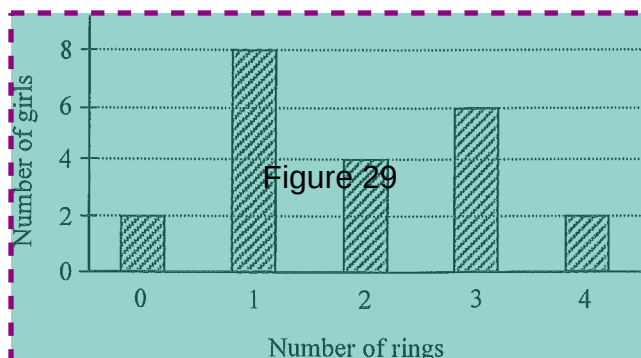
Stem (tens)	Leaf (units)
5	0 5 6 6 8
6	1 4 7 8 8 9
7	3 4 4 6 7 9
8	
9	1

A member is randomly selected from the group. Find the probability that the selected member is not under the age of 74.

- A. 0.2
- B. 0.3
- C. 0.7
- D. 0.8

Label 29 The bar chart below shows the distribution of the numbers of rings owned by the girls in a group. Find the standard deviation of the distribution correct to 2 decimal places.

- A. 1.04
- B. 1.16
- C. 1.19
- D. 2.09



Label 30 Consider the following data:

19 10 12 12 13 13 14 15 16 17 2

If both the mean and the median of the above data are 14, which of the following are true?

- I. $m \geq 14$
- II. $n \leq 16$
- III. $m + n = 30$

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

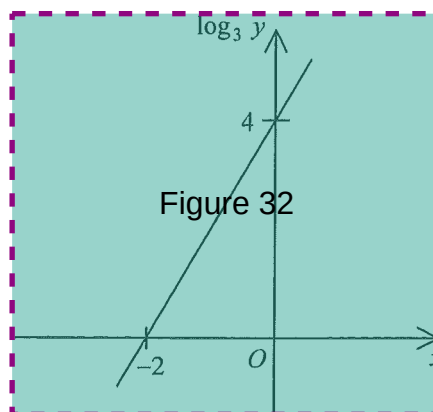
Section B

Label 31 The H.C.F. and the L.C.M. of three expressions are ab^2 and $4a^4b^5c^6$ respectively. If the first expression and the second expression are $2a^2b^4c$ and $4a^4b^2c^6$ respectively, then the third expression is

- A. ab^2
- B. ab^5
- C. $2ab^2c$
- D. $2ab^5c$

Label 32 The graph in the figure shows the linear relation between x and $\log_3 y$. If $y = mn^x$, then $n =$

- A. $\frac{1}{81}$
- B. $\frac{1}{9}$
- C. 9
- D. 81



Label 33 $AD0000002012_{16} =$

- A. $(10)16^{11} + (13)16^{10} + 8210$
- B. $(10)16^{12} + (13)16^{11} + 131360$
- C. $(11)16^{11} + (14)16^{10} + 8210$
- D. $(11)16^{12} + (14)16^{11} + 131360$

Label 34 Let $f(x)$ be a quadratic function. If the coordinates of the vertex of the graph of $y = f(x)$ are $(3, -4)$, which of the following must be true?

- A. The roots of the equation $f(x) = 0$ are integers
- B. The roots of the equation $f(x) - 3 = 0$ are rational numbers
- C. The roots of the equation $f(x) + 4 = 0$ are real numbers
- D. The roots of the equation $f(x) + 5 = 0$ are nonreal numbers

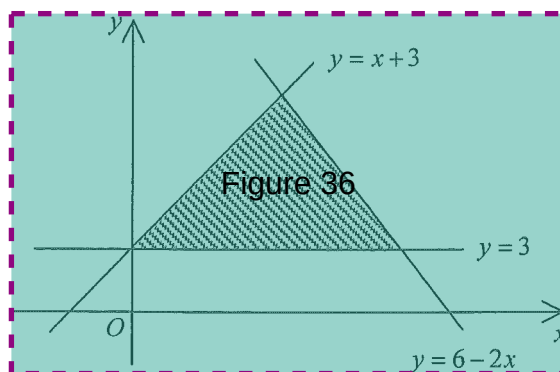
Label 35 $i^3(\beta i - 3) =$

- A. $\beta + 3i$
- B. $\beta - 3i$
- C. $-\beta + 3i$
- D. $-\beta - 3i$

Label 36 The figure shows a shaded region (including the boundary). If (h, k) is a point lying in the shaded region, which of the following are true?

- I. $k \geq 3$
- II. $h - k \geq -3$
- III. $2h + k \leq 6$

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III



Label 37 Let a_n be the n th term of an arithmetic sequence. If $a_{18} = 26$ and $a_{23} = 61$, which of the following are true?

- I. $a_{14} < 0$
- II. $a_1 - a_2 < 0$
- III. $a_1 + a_2 + a_3 + \cdots + a_{27} > 0$

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III